



LITEPAPER V1.1

ROBINHOOD CHAIN - TESTNET FIRST

A FUEL-powered DeFi protocol for recurring ETH and cbBTC rewards while continuously buying and burning FUEL and MORE.

EXPERIMENTAL - UNAUDITED - MOCK ASSET ALPHA

September 2026 | Economics specification and testnet status

Release status

WHAT V1.1 REPRESENTS

This litepaper defines the intended fixed economics and explains what the current public testnet alpha does today. It does not describe the mock deployment as production-ready.

PUBLIC TESTNET ALPHA

- Robinhood Chain Testnet, chain ID 46630.
- Mock FUEL, MORE and cbBTC conversion paths.
- Wallet-connected web interface planned for GitHub Pages.
- Unit, fuzz, integration and stateful invariant testing.

PRODUCTION TARGET

- Verified Robinhood mainnet token and pool contracts.
- TWAP-based FUEL fee pricing.
- Minimum-output and bounded-slippage swap execution.
- Real FUEL, MORE and cbBTC settlement paths.

V1.1 DECISIONS

- All cycle schedules are anchored to 16:00:00 UTC.
- Each reward pool reserves a total 0.25% public settlement budget.
- Fully expired positions may be finalized by anyone after day 14.
- A position closed at the exact reward deadline is excluded from that snapshot.
- Rounding dust rolls into the active cycle after every eligible claim is completed.
- Foundry rewards use cbBTC, not wBTC.

SHAREABLE, NOT A SAFETY GUARANTEE

The alpha is designed for public testing with wallets that hold no real assets. The protocol is unaudited. Test coverage, testnet operation and a polished interface do not prove that production contracts are safe.

Fuel Forge at a glance

ONE ASSET, TRANSFERABLE POSITIONS, RECURRING POOLS

Fuel Forge adds utility to FUEL without creating a new native token. Users lock FUEL, optionally burn more FUEL to increase Forge Power, and compete for recurring externally funded rewards.

8-1,000d

STAKE RANGE

Any whole-day term

5.0x

MAXIMUM POWER

Duration plus burn

8 / 28 / 88

ETH CYCLES

Global recurring pools

288d

FOUNDRY

cbBTC cycle

CORE LOOP

1

STAKE FUEL

Mint a transferable position NFT

2

BUILD POWER

Choose duration and optional burn

3

EARN

Share recurring ETH pools

4

SUPPORT

Fund rewards and buy-and-burn

NO NATIVE FORGE TOKEN

- FUEL is the principal asset users stake and burn.
- Forge Power determines each position's share of ETH rewards and transfers with its NFT.
- ETH and cbBTC are externally funded reward assets.
- FUEL and MORE receive recurring buy-and-burn support from position fees.
- Position NFTs transfer the full economic position without resetting its terms.

Forge Power

LONGER PAYS BETTER + BURN PAYS BETTER

Every normal stake becomes a Forge Position NFT. Principal, initial power, creation time and maturity are fixed at creation. A position cannot be modified or withdrawn early.

FORMULA

$$\text{Power} = \text{Stake} + \text{Duration Bonus} + \text{Burned FUEL}$$

Maximum: 1x base + 1x duration + 3x burn = 5x total

DURATION BONUS

Linear from +0% at 8 days to +100% at 1,000 days.

8 days	0%
88 days	8.1%
365 days	36.0%
730 days	72.8%
1,000 days	100%

BURN BONUS

Burned FUEL adds power 1:1 and is capped at 3x the stake.

0x stake burned	+0%
1x stake burned	+100%
2x stake burned	+200%
3x stake burned	+300% MAX

EXAMPLE

100M FUEL staked for 1,000 days plus 300M FUEL burned:

500M FORGE POWER

100M base + 100M duration bonus + 300M burn bonus. The 300M burn is permanent. The 100M principal follows the maturity and late-finalization rules.

Recurring reward cycles

ELIGIBILITY FREEZES AT THE SCHEDULED TIMESTAMP

The payout clocks are global. Any active position with positive Forge Power at a scheduled close participates, regardless of its chosen term.

Cycle	Asset	Share of 41% bucket	Settlement
8 days	ETH	32%	Batched public trigger
28 days	ETH	27%	Batched public trigger
88 days	ETH	21%	Batched public trigger
288 days - Foundry	cbBTC	20%	Public trigger

SETTLEMENT RULES

- Eligibility is evaluated at the scheduled deadline, even if settlement occurs later.
- A position created or closed at the exact deadline is excluded from that closing cycle.
- Each pool reserves 0.25% total for public settlement execution.
- For 8/28/88-day pools, the 0.25% budget is shared among callers processing bounded batches.
- Rewards are pull-based, follow NFT ownership and never expire.
- After all eligible claims, indivisible rounding dust rolls into the active cycle.

CLOCK ANCHOR

16:00:00 UTC

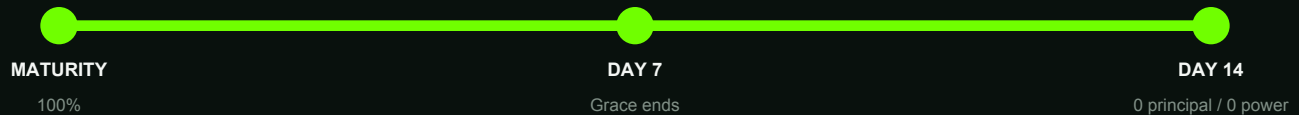
Launch and every future 8/28/88/288-day deadline are derived from this fixed UTC anchor. Delayed settlement cannot move future clocks or capture later-cycle funding.

Position NFTs & maturity

TRANSFERABLE POSITIONS, NO EARLY EXIT

Each stake is represented by a transferable NFT. Ownership controls the remaining principal and every unclaimed reward associated with that position.

LATE-CLAIM TIMELINE



OWNER ACTIONS

- Before maturity: no withdrawal.
- Maturity through day 7: Withdraw returns 100% of remaining principal.
- Day 7 through day 14: Withdraw returns the linearly decayed amount and burns the rest.
- Withdrawing closes and burns the position NFT while preserving past reward rights for its final owner.

PERMISSIONLESS FINALIZATION

At day 14 the economic value is zero. The contract cannot execute by itself, so any wallet or automation may call Finalize expired position.

Finalization sends all remaining principal to the permanent burn address, closes the NFT and records its final owner so earlier rewards remain claimable.

DECAY AND TRANSFERS

- Power remains unchanged during the first seven late days, then decays linearly for seven days.
- Lower power automatically reduces that position's share and increases other positions' relative shares.
- Previously allocated rewards are untouched by later decay or finalization.
- Transfers do not reset principal, power, maturity, decay or claim history.

The 288-Day Foundry

PERMANENT FUEL BURN FOR AN EQUAL SHARE OF CBBTC

The Foundry is separate from normal staking. Users permanently burn FUEL to mint cycle-specific collectible NFTs that share one 288-day cbBTC pool equally.

100M FUEL

COST PER NFT

Permanently burned

cbBTC

REWARD ASSET

Accumulated through cycle

1 NFT = 1

EQUAL SHARE

Unlimited per wallet

HOW IT WORKS

1

BURN

Burn 100M FUEL to mint one Foundry NFT for the active cycle.

2

ACCUMULATE

The Foundry share converts routed ETH into cbBTC.

3

SETTLE

After day 288, a public caller settles the cycle and receives its 0.25% execution reward.

4

CLAIM

Each NFT claims an equal part of the remaining cbBTC. Claims never expire.

AFTER SETTLEMENT

Foundry NFTs never earn from a later cycle. They remain transferable collectibles and carry any unclaimed reward for their original cycle. Once every NFT in a cycle has claimed, any smallest-unit rounding dust rolls into the active Foundry cycle.

Testnet uses deterministic mock cbBTC. Production targets the verified Robinhood Chain cbBTC/WETH market with protected execution.

Protocol fees & buy-and-burn

FIXED ROUTING, PERMISSIONLESS EXECUTION

Creating a normal position requires an ETH fee based only on the ETH market value of staked FUEL. Burned FUEL and duration do not increase the fee.

FEE RULE

0.5% of staked FUEL value in ETH

MIN 0.001 ETH | MAX 0.01 ETH

EVERY 1 ETH OF PROTOCOL REVENUE

41%

REWARDS

34%

FUEL B&B

24%

MORE B&B

1%

DEVELOPMENT

BURN ENGINES

- FUEL and MORE pools target a default 1% of current pool balance per day.
- The daily amount is divided across 144 ten-minute intervals.
- Missed intervals remain executable rather than disappearing.
- Any wallet may execute; 1.5% of the executed interval pays that caller in ETH.
- Production swaps target TWAP-informed minimum output and bounded slippage.
- Purchased FUEL and MORE go directly to permanent burn destinations.

TESTNET NOTE: the public alpha models swaps with fixed mock rates and sink contracts. No real market trade occurs.

Trust model & implementation

FIXED ECONOMICS, EXPLICIT TESTNET LIMITS

HARD-LOCKED ECONOMICS

- 8-1,000-day stake range and power formula.
- 41 / 34 / 24 / 1 revenue routing.
- 32 / 27 / 21 / 20 reward weighting.
- 8 / 28 / 88 / 288-day clocks at 16:00 UTC.
- 100M FUEL Foundry cost.
- Late decay, no early exit and irreversible burns.

PRINCIPAL & REWARDS

- No developer path to withdraw user principal or rewards.
- Pull claims do not expire.
- Failed atomic settlement leaves the transaction reverted for retry.
- Permissionless expiration finalization preserves past reward ownership.
- Rounding dust is conserved through rollover.

PRICING AND SWAPS

The production target uses a short FUEL/WETH TWAP for the 0.5% fee calculation and TWAP-informed minimum outputs for buy-and-burn and ETH-to-cbBTC execution. These controls reduce brief price manipulation and execution risk but cannot eliminate liquidity, oracle or market risk.

The public testnet alpha intentionally substitutes immutable mock rates and mock assets so users can test accounting without real-value tokens.

TESTING STATUS

- Unit and fuzz tests cover formulas, timing, claims and access boundaries.
- Integrated tests exercise the complete fee, Foundry and burn-engine paths.
- Stateful invariants verify conservation through repeated fee routing.
- The protocol is unaudited. Public testnet use is for testing, not proof of safety.

Risks & test participation

EXPERIMENTAL, UNAUDITED, HIGH RISK

The public alpha uses valueless test assets. A future real-asset deployment could cause partial or total loss. Review these risks before interacting with any version of Fuel Forge.

NO EXPECTATION OF PROFIT

Rewards depend on protocol activity and revenue. They are not fixed yield, APY or a guarantee of appreciation.

ASSETS CAN BECOME WORTHLESS

FUEL, MORE, ETH, cbBTC and both NFT types may lose significant value. Buy-and-burn does not guarantee price support.

SMART-CONTRACT AND MARKET RISK

Bugs, exploits, thin liquidity, failed integrations, oracle problems and economic flaws can cause loss.

LOCKS, BURNS AND PRINCIPAL LOSS

There is no early withdrawal. Voluntary and Foundry burns are permanent. Late decay can destroy all principal.

TESTNET IS NOT A SAFETY GUARANTEE

Passing tests and public testnet availability do not guarantee correct or safe production behavior.

PUBLIC ALPHA: USE A NEW TEST-ONLY WALLET AND TEST ETH

Never paste a private key, seed phrase or keystore password into a website, chat or issue report.